OF OSCILLATOR HI

SPECIFICATIONS

1. Frequency Range	LO-2mhz to 22mhz (For Fundamental Xtal) HI-18mhz to 60mhz (Overtone Xtal)
2. RF Output	.2 volts rms into 50 ohms (min)
3. DC Power Required	9 volts at 20ma (operates 6 to 12 volts)
4. Frequency Tolerance with OF and 031300/031310 Crystal	.02%
5. Operating Temperature Range	0 to 50 Degrees Centigrade
6. Frequency Change with 1 volt Supply Change	.001% (max)
7. Output level Change with 1 volt Supply Change	2db (approx)
8. Size	1 1/2 x 1 1/2 x 1 1/4 inches
9. Mounting	4 holes with spacers or fits over 1 1/4 inch chassis hole

LIST OF PARTS

PART	LO VALUE	QTY	HI VALUE	QTY	PART	QTY
Resistor	680 ohm	1	470 ohm	1	Printed Circuit Board	1
					Crystal Socket	1
	100 ohm	1	100 ohm	1	PN5130 Transistor	1
	47K ohm	1	2.2K	1		
			4.7K	1	3/8" spacer	4
					4-40 x 7/8" screw	4
Capacitor	.001mf	1	.001mf	1	#4 hex nut	4
	.01mf	1	.01mf	1	#4 flat washer	4
	470pf	2	100pf	1	#4 lockwasher	4
	220pf	2	220pf	1	Connectors, male for PC board	4
	30pf	1	47pf	1		
			18pf	1	Connectors, female for wire end	4

GENERAL INFORMATION

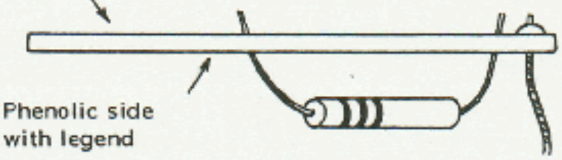
1. The OF oscillator is a resistor/ capacitor circuit providing oscillation over a range of frequencies by merely inserting the desired crystal and without any tuning. The crystal/oscillator combination is intended for general purpose use where close tolerances are not required. The resultant crystal controlled signal will be within .02% of the nominal frequency (600hz at 3000khz or 12khz at 60000khz). The power output is about -10dbm.

2. Unpack the kit and check each part against the parts list. Referring to the parts list and the legend printed on the circuit board, parts can be readily located in their proper position.

PRINTED CIRCUIT WIRING

1. Use only a good grade of rosin-core solder 50/50 or 60/40 tin-lead content. Very little solder is required to make a good connection. Watch for shorts caused by the flow of solder between parts of the foil pattern.

2. Do not overheat the connection. A pencil type iron is ideal for this work. A 25 watt iron is adequate for printed circuit wiring.



3. Components are placed on the phenolic side of the board, with the leads passing through the holes to the foil side. After inserting a part, fan the leads out slightly to prevent the part from falling out until it is soldered.

4. The legend markings on the phenolic side are a guide for placement of parts.

ASSEMBLY

1. The printed circuit board is marked with the value of each part to be inserted in the various holes. A suggested sequence of assembly is given below. After inserting leads through the holes, spread them slightly to hold the part in place until it is soldered. Clip the excess lead length.

2. Insert the .01 and .001 mf disc ceramic capacitors followed by the crystal socket. Install the transistor with its "flat" positioned as shown on the printed circuit board. Position the transistor to stand ¼ inch above the board. The remaining parts (R, C, and L) are selected according to the frequency range from the chart below.

	FREQUENCY RANGE MHZ	R1	R2	R3	R4	C1	C2	C3
OF-LO	3.0 to 15.0	100	680	47K	Not Used	470	470	30
	4.0 to 22.0	100	680	47K	Not Used	220	220	30
OF-HI	18.0 to 29.0	100	470	4.7K	2.2K	220	47	jumper
	2.9 to 60.0	100	470	4.7K	2.2K	100	18	jumper

4. Insert R1 thru R4 and C1 thru C3. Solder all connections and clip any excess component lead on the foil side of the board. Install the desired crystal.

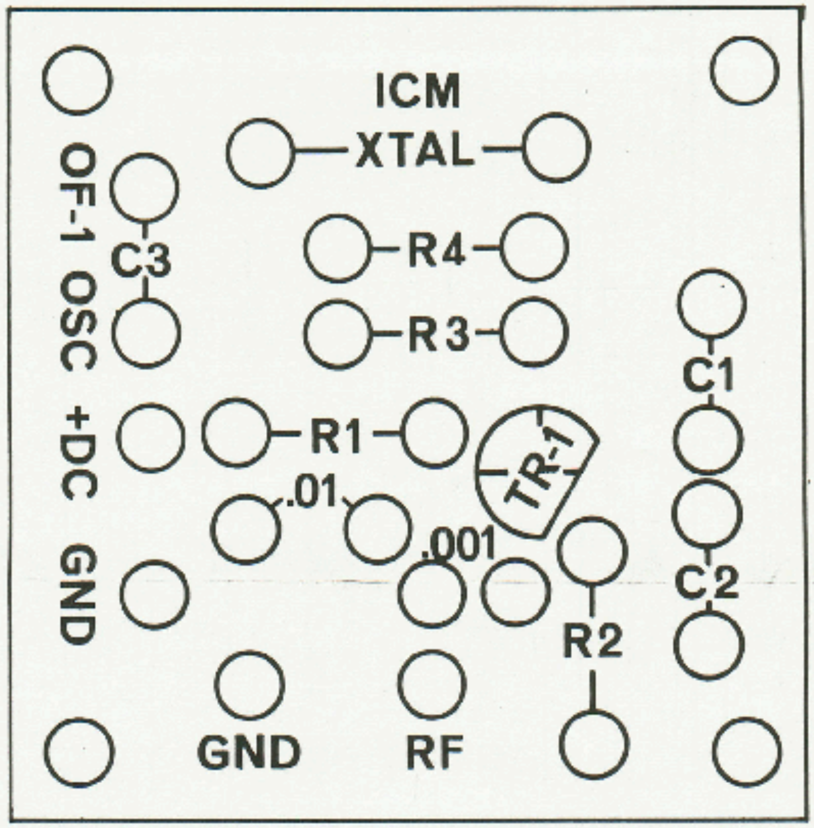
5. Make external connections, applying 9 volts dc to the terminal indicated. Negative 9 volts is connected to the ground terminal.

6. Output is taken at the RF and GND terminals next to the 100 ohm resistor. This is a low impedance source and must be considered in coupling to other equipment.

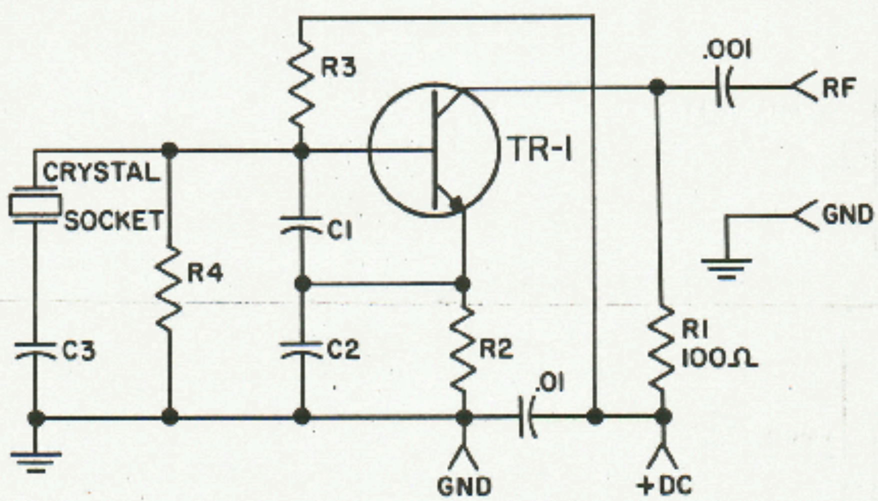
7. Should any difficulty be encountered, look for parts inserted incorrectly, wrong coil range for crystal frequency, or poor solder connection. The oscillator circuit is straight forward and no problem should be had in obtaining proper operation. The transistor has been checked before shipment.

SUGGESTED APPLICATIONS

The following circuits indicate a few of the possible applications for the OF oscillator. For modulation, the oscillator will present approximately a 300 ohm load to the audio source.



OF-1 OSCILLATOR

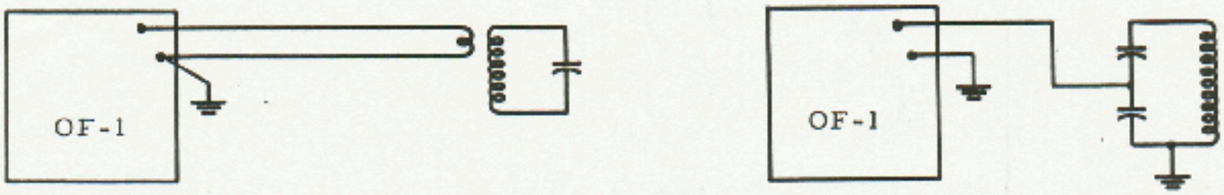


OF-1
LO (FUND)

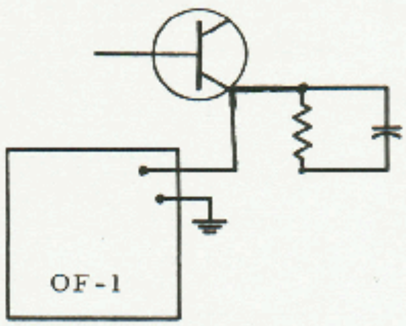
RANGE NO	FREQUENCY MHZ	R4	R2	R3	C1	C2	C3
2	2.0 to 15.0		680	47K	470	470	30
3	4.0 to 22.0		680	47K	220	220	30

OF-1
(3RD OVERTONE)

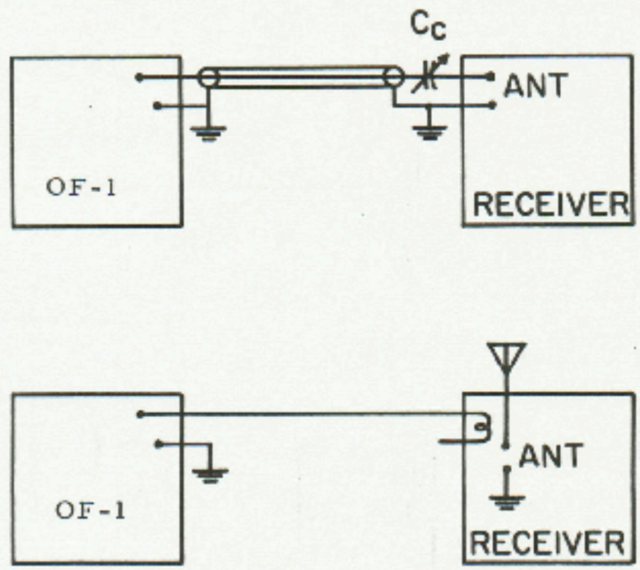
RANGE NO	FREQUENCY MHZ	R4	R2	R3	C1	C2	C3
4	18.0 to 28.0	2200	470	4700	220	47	
5	28.0 to 60.0	2200	470	4700	100	18	



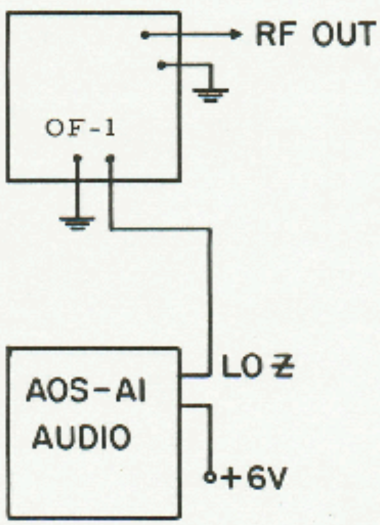
(a) Coupling to High Impedance Circuit.



(b) Emitter Coupling to Mixer.



(c) Coupling to Receiver as Signal Generator. Amount of coupling Adjusted by Cc.



(d) Tone Modulating OF-1 Oscillator.